# The Discriminant-12 Return

Centered Cayley Reciprocity, Pell–Lorentz Geometry, Cyclotomic Closure, and Finite Reduction

Jeremy Ebert

2026

**Abstract**

For $g \in SL_2$ with trace $\tau$, centered part $H = g - \frac{\tau}{2}I$, and Cayley transform $K = (g-I)(g+I)^{-1}$, we prove

$$K = \frac{2}{\tau+2}H, \qquad KH = \frac{\tau-2}{2}I.$$

Thus, in the normalization $\tau = \sigma + 2$, the hyperbolic self-reciprocity condition $K = H^{-1}$ selects $\sigma = 2$, hence trace 4. With the period-$[1,2]$ positive integral normalization this yields

$$g_{12} = \begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix}, \qquad \lambda = 2 + \sqrt{3},$$

with characteristic discriminant 12. We realize $g_{12}$ as multiplication by $\lambda$ on the ramified ideal above 2 in $\mathbb{Q}(\sqrt{3})$, and its centered generator as multiplication by $\sqrt{3}$. The same unit acts on factor-cone null coordinates by $(x, y) \mapsto (\lambda x, \lambda^{-1}y)$ and translates rapidity by $\log \lambda$ on every real shell $xy = n$. The row/column parabola–circle tangent endpoints are its null eigenrays. After adjoining $i$, the operator $\mathcal{Z} = (H_{12} + iI)/2$ is multiplication by $\zeta_{12}$ on the integral cyclotomic ideal $\mathfrak{p}_2\mathcal{O}_{\mathbb{Q}(\zeta_{12})}$. At the ramified prime the cyclotomic residue field is $\mathbb{F}_4$: reduced multiplication by $\zeta_{12}$ has order 3, while the reduced Pell return is Frobenius. Under a compatible additive identification, the reduced Lorentz norm is exactly the finite-field trace, so Boolean XOR parity is $\mathrm{Tr}_{\mathbb{F}_4/\mathbb{F}_2}$. Finally $n = 11$ appears as a distinguished specialization whose Artin action is $T_{11}$, complex conjugation, corresponding to the involution $J$ in the companion representation. No statement about zeta zeros or the Riemann hypothesis is used.

## 1 Introduction

The construction begins before any arithmetic normalization. Centered Cayley reciprocity selects trace 4. A continued-fraction normalization then chooses a distinguished primitive positive integral representative. Its expanding eigenvalue is the fundamental Pell unit $2 + \sqrt{3}$, in the standard quadratic-unit setting of continued fractions and Pell equations [1, 2]. The same operator is subsequently realized on an ideal lattice, as a Lorentz boost on the real factor cone, as a cyclotomic operator after adjoining $i$, and as order-2 and order-3 finite actions after ramified reduction.

The point of the paper is the compatibility of these realizations, rather than any one of the standard background ingredients in isolation. In particular, the integral basis change on the ramified ideal reduces to the Boolean basis change, while the same reduced return becomes Frobenius on $\mathbb{F}_4$. This makes the reduced quadratic parity an exact finite-field trace. The resulting chain is rigid enough to distinguish which identifications are algebraic and which are only representation correspondences.

The logical chain is

$$\boxed{\begin{array}{c} K = H^{-1} \Longleftrightarrow \sigma = 2 \Longrightarrow \tau = 4 \Longrightarrow \Delta = 12 \\ \Longrightarrow \lambda = 2+\sqrt{3} \Longrightarrow g_{12} = \times\lambda \text{ on } \mathfrak{p}_2 \Longrightarrow H_{12} = \times\sqrt{3} \\ \Longrightarrow \mathcal{Z} = \times\zeta_{12} \text{ on } \mathfrak{P}_2 \Longrightarrow \mathbb{F}_4 \text{ reduction.} \end{array}}$$

The real Cone is a parallel carrier of the same Pell return, not a finite quotient. Likewise the Boolean translation group, factor exchange, $U(12) \cong V_4$, and the affine/semilinear groups of $\mathbb{F}_4$ are distinct symmetry layers. Standard background on quadratic forms and ideal classes is used in the usual sense [3, 4]; cyclotomic and finite-field facts are used in their standard forms [5, 6].

The proof is organized in the same dependency order. We first establish the centered Cayley identity, then normalize the selected trace by the period $[1,2]$ return, identify the ramified ideal and Lorentz norm, pass to the real factor cone, complete cyclotomically, and only then reduce to the four-state structures. The $n = 11$ shell is treated last as a specialization rather than as an input.

## 2 Main theorem

**Theorem 2.1** (Discriminant-12 return package)**.** *Let $g \in SL_2(\mathbb{Q})$ have trace $\tau$, assume $g + I$ is invertible, and set*

$$H = g - \frac{\tau}{2}I, \qquad K = (g-I)(g+I)^{-1}, \qquad \tau = \sigma + 2.$$

*Then*

$$K = \frac{2}{\tau+2}H, \qquad KH = \frac{\tau-2}{2}I.$$

*If $g$ is hyperbolic, then*

$$K = H^{-1} \iff \sigma = 2.$$

*At this selected trace $\tau = 4$, impose the period-$[1,2]$ positive integral normalization and define*

$$g_{12} = \begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix}, \qquad H_{12} = g_{12} - 2I.$$

*Then the following hold.*

*(i)* $\chi_{g_{12}}(X) = X^2 - 4X + 1$, $\Delta_{g_{12}} = 12$, *and* $H_{12}^2 = 3I$.

*(ii) On* $\mathfrak{p}_2 = (2, \sqrt{3}-1) = (1+\sqrt{3})$, $g_{12}$ *and* $H_{12}$ *are multiplication by* $2+\sqrt{3}$ *and* $\sqrt{3}$ *respectively.*

*(iii) The primitive form* $q_{12}(x,y) = 2x^2 - 2xy - y^2$ *is preserved by* $g_{12}$ *and satisfies*

$$2q_{12} = U^2 - V^2, \qquad (U,V) = (2x-y, \sqrt{3}\,y).$$

*(iv) In factor-cone null coordinates the same return is*

$$x' = (2+\sqrt{3})x, \qquad y' = (2-\sqrt{3})y,$$

*so $xy$ is invariant and rapidity is translated by* $\log(2+\sqrt{3})$. *The factor-parabola tangent endpoints are the two null eigenrays.*

*(v) If $K_{12} = \mathbb{Q}(\zeta_{12})$ and $\mathfrak{P}_2 = \mathfrak{p}_2 \mathcal{O}_{K_{12}}$, then*

$$\mathcal{Z} = \frac{H_{12} + iI}{2} = \times\zeta_{12}$$

*on $\mathfrak{P}_2$, and $[\mathcal{O}_{K_{12}} : \mathbb{Z}[\sqrt{3}, i]] = 4$.*

*(vi) $\mathcal{O}_{K_{12}}/\mathfrak{P}_2 \cong \mathbb{F}_4$. Reduced $\mathcal{Z}$ is multiplication by an element $\omega$ of order 3, while reduced $g_{12}$ is Frobenius. These generate $GL_2(\mathbb{F}_2) \cong S_3$.*

# 3 Centered Cayley reciprocity

**Lemma 3.1.** *For $g \in SL_2$ with trace $\tau$,*

$$H^2 = \left(\frac{\tau^2}{4} - 1\right) I.$$

*Proof.* Cayley–Hamilton gives $g^2 - \tau g + I = 0$. Substituting $H = g - \tau I/2$ gives the result. □

**Proposition 3.2.** *If $g + I$ is invertible, then*

$$K = \frac{2}{\tau + 2} H, \qquad KH = \frac{\tau - 2}{2} I.$$

*Proof.* Cayley–Hamilton gives

$$(g + I)((\tau + 1)I - g) = (\tau + 2)I,$$

whence

$$(g + I)^{-1} = \frac{(\tau + 1)I - g}{\tau + 2}.$$

Therefore

$$K = \frac{2g - \tau I}{\tau + 2} = \frac{2}{\tau + 2} H,$$

and multiplication by $H$ gives the second identity. □

**Corollary 3.3.** *If $g$ is hyperbolic and $\tau = \sigma + 2$, then*

$$K = H^{-1} \iff \sigma = 2.$$

*Proof.* Hyperbolicity makes $H$ invertible, so $K = H^{-1}$ iff $KH = I$, which is $(\tau - 2)/2 = 1$. □

# 4 The primitive integral return and the ramified ideal

Let

$$A_a = \begin{pmatrix} a & 1 \\ 1 & 0 \end{pmatrix}.$$

The period $[1, 2]$ gives

$$g_{12} = A_1 A_2 = \begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix}.$$

The Cayley condition selects the trace-4 hyperbolic class; the period/positivity normalization selects this representative. Directly,

$$\det g_{12} = 1, \quad \operatorname{tr} g_{12} = 4, \quad \chi_{g_{12}}(X) = X^2 - 4X + 1,$$

so the eigenvalues are $2 \pm \sqrt{3}$ and the characteristic discriminant is 12. For

$$H_{12} = g_{12} - 2I = \begin{pmatrix} 1 & 1 \\ 2 & -1 \end{pmatrix},$$

one has $H_{12}^2 = 3I$.

Let

$$f_1 = 2, \qquad f_2 = \sqrt{3} - 1, \qquad \mathfrak{p}_2 = (f_1, f_2).$$

Then

$$\sqrt{3} f_1 = f_1 + 2f_2, \qquad \sqrt{3} f_2 = f_1 - f_2,$$

so

$$[\times\sqrt{3}]_f = H_{12}, \qquad [\times(2+\sqrt{3})]_f = g_{12}.$$

This is a direct ideal-lattice realization in the quadratic field; no classification theorem is needed for the matrix computation itself.

A second natural basis is

$$e_1 = 1 + \sqrt{3}, \qquad e_2 = 3 + \sqrt{3} = \sqrt{3} e_1.$$

Since $e_1 = f_1 + f_2$ and $e_2 = 2f_1 + f_2$,

$$C_{\text{basis}} = \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix}, \qquad C_{\text{basis}}^{-1} g_{12} C_{\text{basis}} = \begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix}.$$

Modulo 2, this integral basis change will become the Boolean relabeling.

## 5 Pell–Lorentz and factor-cone realizations

For

$$\alpha = x f_1 + y f_2 = (2x - y) + y\sqrt{3},$$
$$N_{F/\mathbb{Q}}(\alpha) = (2x - y)^2 - 3y^2 = 2q_{12}(x, y),$$

where

$$q_{12}(x, y) = 2x^2 - 2xy - y^2.$$

Thus, with

$$U = 2x - y, \qquad V = \sqrt{3}\, y,$$
$$2q_{12} = U^2 - V^2.$$

Multiplication by $2 + \sqrt{3}$ is the Lorentz boost

$$\begin{pmatrix} U' \\ V' \end{pmatrix} = \begin{pmatrix} 2 & \sqrt{3} \\ \sqrt{3} & 2 \end{pmatrix} \begin{pmatrix} U \\ V \end{pmatrix}.$$

Now introduce Paper-A factor-cone coordinates

$$X = \frac{x-y}{2}, \qquad Y = \sqrt{xy}, \qquad T = \frac{x+y}{2}.$$

Then

$$T^2 - X^2 - Y^2 = 0, \qquad x = T + X, \qquad y = T - X.$$

Consequently

$$\boxed{x' = \lambda x, \qquad y' = \lambda^{-1} y,} \qquad \lambda = 2 + \sqrt{3},$$

and $xy = n$ is invariant. If

$$x = \sqrt{n} e^{s}, \qquad y = \sqrt{n} e^{-s},$$

then

$$\boxed{s' = s + \log(2 + \sqrt{3}).}$$

Factor exchange sends $s \mapsto -s$ and reverses the boost.

For fixed factor level $u > 0$, the row parabola is

$$Y^2 = u^2 + 2uX$$

with vertex $(-u/2, 0)$, while its reflected column parabola has vertex $(u/2, 0)$. A fixed-$T$ Cone slice is

$$X^2 + Y^2 = T^2.$$

Tangency at the vertices occurs exactly for

$$T = u/2.$$

Hence the side-view tangent points are

$$(T, X) = \left(\frac{u}{2}, \pm\frac{u}{2}\right),$$

which are precisely the two null eigenrays of the discriminant-12 boost. Figure 1 displays the complete half-step fixed-$T$ family through $x + y = 12$ and emphasizes the exact $u = 5, 6, 7$ tangencies. The figure is illustrative of the theorem above; the null-eigenray statement does not depend on the plotted discretization.

## 6 The mod-12 unit shell

The units modulo 12 are

$$U(12) = \{1, 5, 7, 11\} \cong V_4.$$

For $r \in U(12)$, the factor pairs $(r, 1)$ and $(1, r)$ have

$$T_r = \frac{r+1}{2}, \qquad X_r = \pm\frac{r-1}{2}, \qquad Y_r = \sqrt{r}.$$

Thus the four labels occupy $T = 1, 3, 4, 6$, and the outer $r = 11$ pair lies on $x + y = 12$. The standard cyclotomic action is

$$\mathrm{Gal}(\mathbb{Q}(\zeta_{12})/\mathbb{Q}) \cong U(12), \qquad T_r : \zeta_{12} \mapsto \zeta_{12}^{r},$$

as usual for cyclotomic fields [5]. The geometry carries the labels but does not define their modular group law.

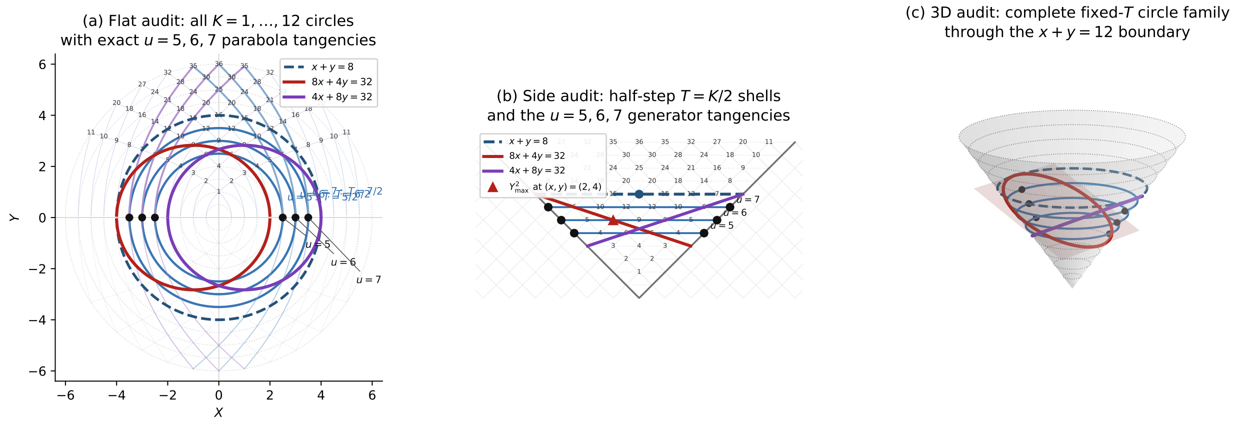


Figure 1: Parabola–circle tangency and the discriminant-12 null eigenrays. The flat projection, side view, and full Cone show the fixed-$T$ family $T = K/2$ for $K = 1, \ldots, 12$. The emphasized $u = 5, 6, 7$ row/column parabolas are tangent at $(T, X) = (u/2, \pm u/2)$, precisely the two null eigendirections of the Pell boost.

# 7 Cyclotomic completion on an integral ideal

Let

$$K_{12} = \mathbb{Q}(\zeta_{12}) = \mathbb{Q}(\sqrt{3}, i), \qquad \zeta = \frac{\sqrt{3} + i}{2}.$$

Since $H_{12} = \times\sqrt{3}$,

$$\boxed{\mathcal{Z} = \frac{H_{12} + iI}{2} = \times\zeta.}$$

The correct integral carrier is

$$\mathfrak{P}_2 = \mathfrak{p}_2 \mathcal{O}_{K_{12}}.$$

For $\alpha = 1 + \sqrt{3}$,

$$\{\alpha, \alpha\zeta, \alpha\zeta^2, \alpha\zeta^3\}$$

is a $\mathbb{Z}$-basis of $\mathfrak{P}_2$. Since

$$\Phi_{12}(X) = X^4 - X^2 + 1, \qquad \zeta^4 = \zeta^2 - 1,$$

multiplication by $\zeta$ has matrix

$$\boxed{\begin{pmatrix} 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}}.$$

Thus the half in $(H_{12} + iI)/2$ is a coordinate effect, not a failure of integrality on the correct lattice.

The full ring of integers is $\mathcal{O}_{K_{12}} = \mathbb{Z}[\zeta_{12}]$, whereas $\mathcal{O}_0 = \mathbb{Z}[\sqrt{3}, i]$ is a suborder of index

$$\boxed{[\mathcal{O}_{K_{12}} : \mathcal{O}_0] = 4.}$$

Equivalently,

$$\operatorname{disc}(\mathcal{O}_0) = 2304, \qquad \operatorname{disc}(\mathcal{O}_{K_{12}}) = 144.$$

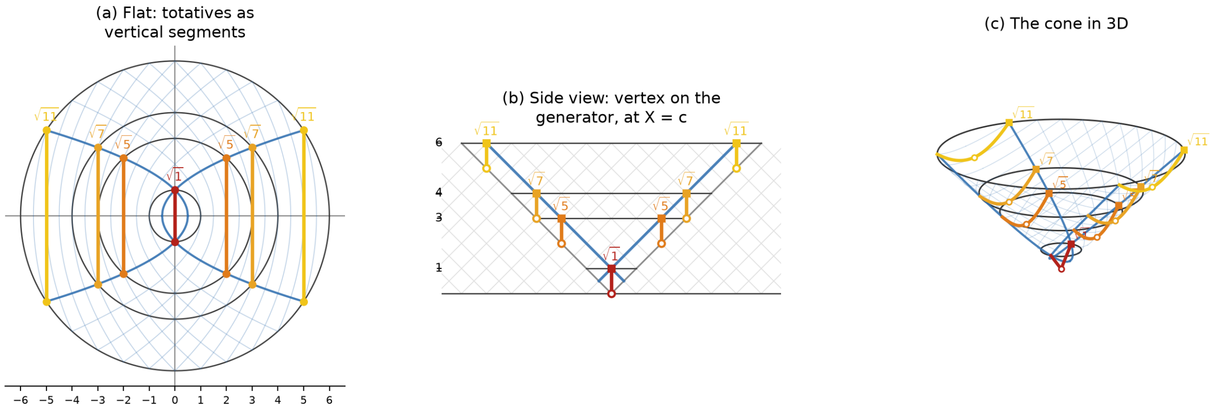


Figure 2: The mod-12 unit labels in factor-cone coordinates. The outer $r = 11$ pair is $(11, 1)$ and $(1, 11)$ on $x + y = 12$. The geometric placement realizes the four labels but is not itself the multiplication law of $U(12)$.

These are standard algebraic-number-theoretic notions of integral closure and field discriminant [4, 5]. The additive quotient $\mathcal{O}_{K_{12}}/\mathcal{O}_0 \cong (\mathbb{F}_2)^2$ records integral-closure gluing; it is not the residue field below.

# 8 Ramified reduction, Frobenius, and XOR trace

Modulo 2 in the ramified basis,

$$\bar{g}_{12} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad \bar{q}_{12}(x, y) = y.$$

Also

$$C_{\text{basis}} \equiv \Phi = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \pmod 2,$$

and therefore

$$\Phi \bar{g}_{12} \Phi^{-1} = P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Thus the Boolean relabeling is the reduction of an integral ideal basis change. In Boolean coordinates $(\epsilon_1, \epsilon_2)$,

$$\bar{q}_{12} = \epsilon_1 \oplus \epsilon_2.$$

Together with Boolean translations, factor exchange gives $V_4^{\text{tr}} \rtimes C_2 \cong D_8$.

The cyclotomic residue field is a different four-state ring:

$$\boxed{\mathcal{O}_{K_{12}}/\mathfrak{P}_2 \cong \mathbb{F}_4.}$$

Let $\omega = \bar{\zeta}$. Then

$$\omega^2 + \omega + 1 = 0, \qquad \omega^3 = 1.$$

In the additive basis $(1, \omega)$,

$$[\times\omega] = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \qquad [\text{Fr}] = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Hence

$$\boxed{[\mathrm{Fr}] = \bar{g}_{12}}, \qquad \mathrm{Fr}(\times\omega)\,\mathrm{Fr}^{-1} = (\times\omega)^{-1},$$

so

$$\langle \times\omega, \mathrm{Fr}\rangle = GL_2(\mathbb{F}_2) \cong S_3.$$

With translations,

$$\mathrm{AGL}(1,4) \cong A_4, \qquad \mathrm{A\Gamma L}(1,4) \cong S_4.$$

The Frobenius and trace statements here are the standard degree-2 finite-field constructions specialized to $\mathbb{F}_4$ [6].

### 8.1 Compatible additive identification and finite-field trace

There is a particularly useful $\mathbb{F}_2$-linear identification between the ramified four-state additive group and $\mathbb{F}_4$. Send

$$\boxed{f_1 \longmapsto 1, \qquad f_2 \longmapsto \omega^2.}$$

Because $e_1 = f_1 + f_2$ and $e_2 = f_2$ modulo 2, this gives

$$e_1 \longmapsto 1 + \omega^2 = \omega, \qquad e_2 \longmapsto \omega^2.$$

Thus the principal/Pell basis corresponds to $(\omega, \omega^2)$, a basis interchanged by Frobenius. Consequently the same involution is represented as the transvection $\bar{g}_{12}$ in the ramified basis and as factor exchange $P$ in the principal/Boolean basis.

The field trace satisfies

$$\mathrm{Tr}_{\mathbb{F}_4/\mathbb{F}_2}(1) = 0, \qquad \mathrm{Tr}_{\mathbb{F}_4/\mathbb{F}_2}(\omega) = \mathrm{Tr}_{\mathbb{F}_4/\mathbb{F}_2}(\omega^2) = 1.$$

Hence in the ramified basis

$$\mathrm{Tr}_{\mathbb{F}_4/\mathbb{F}_2}(x + y\omega^2) = y = \bar{q}_{12}(x, y),$$

while in the Boolean/Pell basis

$$\boxed{\mathrm{Tr}_{\mathbb{F}_4/\mathbb{F}_2}(\epsilon_1\omega + \epsilon_2\omega^2) = \epsilon_1 \oplus \epsilon_2.}$$

Therefore, under this compatible additive identification,

$$\boxed{\bar{q}_{12} = \mathrm{Tr}_{\mathbb{F}_4/\mathbb{F}_2},}$$

so the Boolean XOR parity is exactly the finite-field trace.

*Remark* 8.1 (Ring guardrail). This is an additive $\mathbb{F}_2$-linear identification, not a ring isomorphism. The ramified quotient $\mathfrak{p}_2/2\mathfrak{p}_2$ is nonreduced, whereas $\mathcal{O}_{K_{12}}/\mathfrak{P}_2 \cong \mathbb{F}_4$ is a field.

## 9 The distinguished $n = 11$ specialization

The construction above does not require $n = 11$. It enters afterward as a distinguished specialization. Geometrically,

$$(11, 1), (1, 11)$$

lie on $xy = 11$ and $x + y = 12$, with

$$(X, Y, T) = (\pm 5, \sqrt{11}, 6).$$

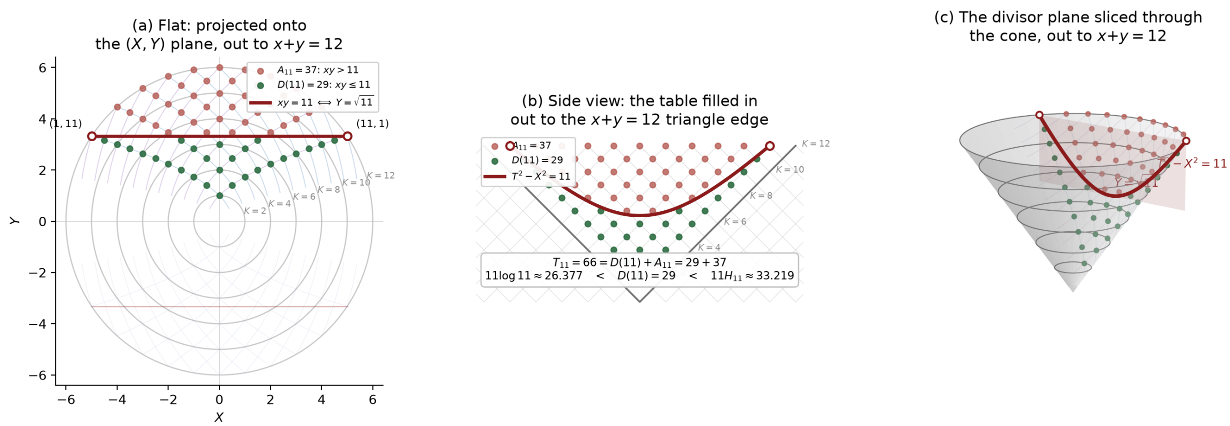


Figure 3: The $n = 11$ divisor shell in the flat, side, and three-dimensional Cone views. The figure relates the shell to the divisor summatory count and the comparison quantities used in Paper A, while preserving the distinguished endpoints $(11, 1)$ and $(1, 11)$ at $(X, Y, T) = (\pm 5, \sqrt{11}, 6)$.

Figure 3 places this shell in the same three views used for the Cone geometry and overlays the corresponding divisor summatory quantities. Its purpose is explanatory: the arithmetic specialization below is established independently of the visualization.

Arithmetically,

$$N_{F/\mathbb{Q}}(1 + 2\sqrt{3}) = -11.$$

Let

$$\mathfrak{p}_{11} = (1 + 2\sqrt{3}).$$

Since

$$(1 + \sqrt{3})(1 + 2\sqrt{3}) = 7 + 3\sqrt{3}$$

is totally positive,

$$\boxed{[\mathfrak{p}_{11}] = [\mathfrak{p}_2]}$$

in the narrow ideal class group. For the narrow Hilbert class field $K_{12} = \mathbb{Q}(\zeta_{12})$ of $F = \mathbb{Q}(\sqrt{3})$,

$$\boxed{\mathrm{Art}_{K_{12}/F}(\mathfrak{p}_{11}) = T_{11}.}$$

The use of the narrow class group and its Artin reciprocity map is standard class-field theory terminology [4, 3]. Here $T_{11} \in \mathrm{Gal}(K_{12}/F)$ because

$$T_{11}(\sqrt{3}) = \sqrt{3},$$

and, since $11 \equiv -1 \pmod{12}$,

$$T_{11}(\zeta_{12}) = \zeta_{12}^{11} = \zeta_{12}^{-1} = \overline{\zeta_{12}}.$$

Thus $T_{11}$ is complex conjugation over $F$.

In the companion operator representation this involution corresponds to $J$:

$$\boxed{T_{11} \longleftrightarrow J.}$$

This is a representation correspondence, not a literal equality between a field automorphism and an operator in another category. The specialization chain is

$$\boxed{11 \longrightarrow \mathfrak{p}_{11} \longrightarrow [\mathfrak{p}_{11}] = [\mathfrak{p}_2] \longrightarrow T_{11} \longleftrightarrow J.}$$

## 10 Symmetry layers and scope

The finite groups appearing above have different roles:

$$\begin{array}{c|c} U(12) \cong V_4 & \text{cyclotomic Galois labels,} \\ V_4^{\mathrm{tr}} & \text{Boolean translations,} \\ V_4^{\mathrm{tr}} \rtimes C_2 \cong D_8 & \text{translations plus factor exchange,} \\ \mathbb{F}_4^{\times} \cong C_3 & \text{nonzero residue-field multiplication,} \\ \mathrm{AGL}(1,4) \cong A_4 & \text{affine field symmetry,} \\ \mathrm{A\Gamma L}(1,4) \cong S_4 & \text{affine semilinear symmetry.} \end{array}$$

They are compatible layers, not interchangeable groups.

A separate analytic companion may realize the same Pell unit through a certified operator crossing. That realization is not used here. In particular, the present paper does not infer analytic zero locations from the finite reductions, the class-field description, or the Cone geometry. It proves no statement about zeta-zero locations and makes no claim concerning the Riemann hypothesis.

## 11 Conclusion

Centered Cayley reciprocity selects trace 4 before arithmetic data enter. The positive period-$[1, 2]$ normalization then exposes the Pell unit $2 + \sqrt{3}$ and fixes the integral representative $g_{12}$. The subsequent steps do not introduce unrelated structures: they transport the same return through compatible carriers. On the ramified ideal it is multiplication by the Pell unit; after centering it is multiplication by $\sqrt{3}$; on the real factor cone it is a Lorentz boost with null eigenrays equal to the exact parabola–circle tangent endpoints; and after cyclotomic completion it becomes multiplication by $\zeta_{12}$ on an integral rank-4 ideal.

The finite reduction then separates the cyclotomic order-3 action from the order-2 Frobenius action. Under the basis identification inherited from the integral ideal, the reduced quadratic parity is the field trace

$$\bar{q}_{12} = \mathrm{Tr}_{\mathbb{F}_4/\mathbb{F}_2},$$

so Boolean XOR is not an added convention but the trace of the same four-state carrier. The $n = 11$ shell finally supplies a distinguished specialization in which the outer $x + y = 12$ geometry, the ramified narrow class, cyclotomic complex conjugation $T_{11}$, and the companion involution $J$ meet without being used to define the theory.

The resulting package is therefore best read as an exact compatibility statement across real, integral, cyclotomic, and finite representations of a single discriminant-12 return. Any further analytic application is a separate problem and requires additional hypotheses and proofs.

## References


[1] G. H. Hardy and E. M. Wright, *An Introduction to the Theory of Numbers*, 6th ed., Oxford University Press, 2008.

[2] J.-P. Serre, *A Course in Arithmetic*, Springer, 1973.

[3] D. A. Cox, *Primes of the Form $x^2+ny^2$: Fermat, Class Field Theory, and Complex Multiplication*, 2nd ed., Wiley, 2013.

[4] J. Neukirch, *Algebraic Number Theory*, Springer, 1999.

[5] L. C. Washington, *Introduction to Cyclotomic Fields*, 2nd ed., Springer, 1997.

[6] R. Lidl and H. Niederreiter, *Finite Fields*, 2nd ed., Cambridge University Press, 1997.